

# National Academy of Science and Technology

(Affiliated to Pokhara University)

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## Pre-University Examination

Level: Bachelor

Semester : VII\_Fall

Year : 2025

Program: B.E. Computer

F.M. : 100

Course: Cloud Computing and Virtualization

P. M. : 45

Time : 3 Hrs.

*Candidates are required to give their answer in their own words as far as practicable. The figures in the margin indicate full marks.*

**Attempt all questions.**

1. a) Explain NIST Cloud Reference Architecture in detail. List down the design principles of cloud architecture. 8  
b) Compare and contrast the three primary Cloud Service Models. 7
2. a) Explain Jericho Cloud Cube Model. How does this multi-dimensional framework help organizations assess the security and boundary implications of different cloud deployment scenarios? 7  
b) Define Virtualization. Explain the different Virtualization Levels. 8
3. a) Differentiate between Type 1 and Type 2 Hypervisors. In a high-performance production environment, why is a "bare-metal" hypervisor preferred over a hosted one? Give Reasons. 8  
b) Discuss the architecture of a Virtual Private Cloud (VPC). How do subnets, internal gateways, and Multihoming contribute to network cost optimization and security? 7
4. a) Discuss the concepts of Data Redundancy and Elastic Storage. How do these features ensure high data durability and accessibility in a Distributed File System? 7  
b) Define Database as a Service (DBaaS) and explain its advantages over managing a self-hosted database on a virtual machine 8
5. a) Define Scaling and describe the scaling Strategy that utilizes both Vertical and Horizontal Scaling. 8  
b) Discuss the role of Load Balancing in achieving High Availability. How does a cloud provider manage redundancy to prevent single points of failure? 7

6. a) Evaluate the Dimensions of Cloud Security. How do Identity and Access Management (IAM) and Single Sign-On (SSO) provide a robust defense-in-depth against advanced cloud security threats? 3

b) What are the primary Challenges of Cloud Security when dealing with multi-tenant environments? 7

7. Write short notes on: (Any two) 2x5

a) AWS

b) Cloud Deployment Models.

c) Security as a Service

**Good Luck!!!**